Applied Chemistry (IUPAC) approved the official name and symbols for four newly discovered elements : Nihonium (Nh), Moscovium (Mc), Tennessine (Ts), and Oganesson (Og), respectively for element 113, 115, 117, and 118. Among them, the most recently discovered element is element with atomic number 117 (Tennessine), the discovery of which was officially announced in April, 2010.

11. Which of the following is a fundamental element :

(a) Sand (b) Diamond
(c) Marble (d) Sugar

U.P.P.C.S. (Pre) 1995

Ans. (b)

Chemically diamond is the purest form of Carbon (crystal structure). Hence, it is a fundamental element. Sand is basically made up of Silicon and Oxygen, Marble is made up of Calcium, Carbon and Oxygen while sugar is mainly the compound of Carbon, Hydrogen and Oxygen.

12. The chemical composition of diamond is?

(a) Carbon (b) Nitrogen
(c) Nickel (d) Zinc

Uttarakhand P.C.S. (Pre) 2007

Ans. (a)

See the explanation of above question.

13. Which of the following is not a type of element?

(a) Metals (b) Nonmetals
(c) Gases (d) Metalloids
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Elements of the periodic table are grouped as metals, metalloids (or semimetals) and nonmetals. Some Gases (hydrogen, helium, nitrogen, oxygen, fluorine, neon, chlorine, argon, krypton, xenon and radon) are elements & categorised as nonmetals, but most of the gases are compounds.

14. Which of the following elements was first produced artificially?

(a) Neptunium (b) Plutonium
(c) Francium (d) Technetium

Jharkhand P.C.S. (Pre) 2013

Ans. (d)

Technetium (Tc) is a chemical element with atomic number 43. It is a radioactive, silvery-gray transition metal. It is the first artificially created element.Technetium was isolated by Carlo Perrier and Emilio Segre in 1937.

15. Which of the following is the first artificially prepared element ?

(a) Te (b) Tc
(c) Th (d) Tl

70th B.P.S.C. (Pre) 2024

Ans. (b)

See the explanation of above question.

Metals, Minerals, Ores : Properties, Uses

Notes

- Earth crust is the main source of elements.
- Among non-metals - **Oxygen** and then Silicon and among metals - **Aluminium** is found in abundant quantity in the earth crust.
- In nature, metals are found in two states-

(i) Free state
(ii) Combined state

- In nature, the occurrence of any metal depends on its chemical reactivity.
- Less reactive metals as gold and platinum occurred in a free state.
- More reactive metals are found in combined state.
- Some metals as copper, silver and iron are found in both states i.e. free and combined state.

Minerals :

- The state in which metals and their compounds occur in earth's crust is known as minerals.

Ores :

- Minerals from which economically beneficial metals or gems can be extracted are known as ores.
- Generally, metals occur in the state of oxides, sulphides, carbonates, halides and sulphates.

Minerals and Ores of Some Metals

Metal	Mineral/Ores	Composition of Mineral/Ores
Sodium (Na)	Rock Salt Trona Chilli Saltpetre Borex Glauberite Glauber's Salt	$NaCl$ $Na_3(HCO_3)(CO_3)\ 2H_2O$ $NaNO_3$ $Na_2B_4O_7.10H_2O$ $Na_2Ca(SO_4)_2$ $Na_2\ SO_4.10H_2O$
Potassium (K)	Sylvine Carnallite Schoenite (Picromerite)	KCl $KCl.MgCl_2.6H_2O$ $K_2Mg(SO_4)_2.6H_2O$

Magnesium (Mg)	Magnesite Dolomite Carnallite Kieserite Epsom Salt (Epsomite)	$MgCO_3$ $MgCO_3.CaCO_3$ $KCl.MgCl_2.6H_2O$ $MgSO_4.H_2O$ $MgSO_4.7H_2O$
Calcium (Ca)	Limestone Calcite Gypsum Fluorspar	$CaCO_3$ $CaCO_3$ $CaSO_4.2H_2O$ CaF_2
Aluminium (Al)	Bauxite Cryolite Corundum Diaspore	$Al_2O_3.2H_2O$ Na_3AlF_6 Al_2O_3 $AlO(OH)$
Tin (Sn)	Cassiterite (Tinstone)	SnO_2
Lead (Pb)	Galena Cerussite Matlockite	PbS $PbCO_3$ $PbFCl$
Copper (Cu)	Chalcopyrite (Copper Pyrite) Chalcocite Cuprite Malachite Azurite	$CuFeS_2$ Cu_2S Cu_2O $Cu_2(CO_3)(OH)_2$ $Cu_3(CO_3)_2(OH)_2$
Silver (Ag)	Native Silver Argentite Cerargyrite	Ag Ag_2S (Silver Glance) AgCl (Horn Silver)
Zinc (Zn)	Zinc Blende Franklinite Calamine Zincite	ZnS (Black Jack) $ZnFe_2^{3+}O_4$ $ZnCO_3$ ZnO (Red Zinc)
Mercury (Hg)	Cinnabar	HgS
Manganese (Mn)	Pyrolusite Manganite Hausmannite	MnO_2 $MnO(OH)$ Mn_3O_4
Iron (Fe)	Magnetite Haematite Limonite Siderite Iron Pyrites	Fe_3O_4 (Load stone or Magnetic oxide of iron) Fe_2O_3 $FeO(OH).nH_2O$ $FeCO_3$ (Spathic Iron) FeS_2

Question Bank

1. The metallurgical process in which a metal is obtained in a fused state is called :

(a) smelting (b) roasting
(c) calcination (d) froth floatation
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Metal is obtained at high temperature by the reduction of ore in the process of smelting. The metal is obtained in a fused state in this process.

2. Which of the following is the electron configuration of a metallic element?

(a) 2, 8 (b) 2, 8, 7
(c) 2, 8, 8 (d) 2, 8, 8, 2

45th B.P.S.C. (Pre) 2001

Ans. (d)

Calcium is a dull, grey, solid element with a silver appearance which exists in the solid state. It has a high melting point (1115k) and boiling point (1757 K). All these features make it related to the metals. The valence electron configuration of Calcium is 2,8,8,2. Hence, it has a tendency to lose two electrons to get a noble gas configuration. Since it can lose electrons, it can be used in ionic bonding and can form ionic compounds. Like other metals, Calcium also reacts vigorously with dilute acids like hydrochloric acid and produce large amounts of heat, forms Calcium Chloride ($CaCl_2$) and Hydrogen gas. All these properties of Calcium prove that it is a metal.

3. Which of the following is not an electrophile?

(a) Na^+ (b) BF_3
(c) H^+ (d) All of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (a)

Na^+ (Sodium ion) is not considered a true electrophile. While Na^+ is a positively charged ion, it already has a complete or stable octet, making it less likely to accept electrons. Electrophiles are species that accept electron and are often electron-deficient. While Na^+ does have a positive charge, it doesn't have readily available vacant orbitals of lower energy to accept electrons to function as a true electrophile. Boron trifluoride (BF_3) and hydrogen ion (H^+) are true electrophiles.

4. Which is the most reactive metal ?

(a) Sodium (b) Calcium

(c) Iron (d) Potassium

56th to 59th B.P.S.C. (Pre) 2015

Uttarakhand P.C.S. (Pre) 2024

Ans. (d)

In the given options, according to the reactivity series, Potassium (K) is the most reactive metal. It reacts vigorously with air and water, producing a large amount of heat. The ionization energy of potassium is lower than that of sodium because its atomic size is larger. Metals like potassium and sodium react rapidly with cold water. The reactions of potassium and sodium are vigorous and exothermic because the hydrogen released immediately ignites. Calcium reacts with water somewhat slowly. Magnesium does not react with cold water; however, it reacts with hot water to produce magnesium hydroxide and hydrogen gas. Metals like aluminium, iron, and zinc do not react with either cold water or hot water; instead, they react with steam to give metal oxides and hydrogen. The metals at the top of the reactivity series (K, Na, Ca, Mg, and Al) are so reactive that they are never found in a free state.

Reactivity series of metals :

K	Potassium	Most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
Zn	Zinc	Reactivity decreases
Fe	Iron	
Pb	Lead	
[H]	[Hydrogen]	
Cu	Copper	
Hg	Mercury	
Ag	Silver	
Au	Gold	Least reactive

5. Food cans are coated with Tin instead of Zinc, because:

(a) Zinc is expensive than Tin.

(b) The melting point of Zinc is higher than Tin.

(c) Zinc is more reactive than Tin.

(d) Zinc is less reactive than Tin.

Uttarakhand P.C.S. (Pre) 2024

Ans. (c)

Zinc (Zn) is a more reactive metal than tin (Sn). This means that zinc is more likely to react with other substances, including the organic acids present in food. If zinc were used to coat food cans, it could react with the food, potentially dissolving into the food and making it unsafe to consume. Tin is a less reactive metal, meaning it is less likely to react with the food contents. This makes tin a safer and more suitable choice for coating food cans, as it helps prevent the food from being contaminated or spoiled by the can itself.

6. The paramagnetic theory of magnetism applies to :

(a) nickel (b) mercury

(c) iron (d) platinum

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

Paramagnetism is a form of magnetism in which some material is weakly attracted by an externally applied magnetic field and creates an internal, induced magnetic field in the direction of the applied magnetic field. Among the given metals, paramagnetic theory of magnetism applies to platinum.

7. Which among the following pair is an example of paramagnetic and ferromagnetic substances?

(a) H_2O and Fe (b) Cu^{2+} and Gd

(c) Cr^{3+} and MnO (d) Fe and Cu^{2+}

Chhattisgarh P.C.S. (Pre) 2024

Ans. (b)

Paramagnetism, ferromagnetism and diamagnetism refer to different magnetic properties of matter. Ferromagnetic substances are completely attracted towards a magnetic field and retain their own magnetic properties even in the absence of an applied magnetic field. The spin of the unpaired electrons present in them is aligned, resulting in a strong magnetic moment. Paramagnetic substances are partially attracted towards a magnetic field and lose their own magnetic properties in the absence of an applied magnetic field. They have unpaired electrons, but the spin of the electrons present in them is randomly oriented. Diamagnetic substances are weakly repelled by a magnetic field. Their all electrons are paired, resulting in negative magnetic susceptibility. Among the given substances H_2O (water) is diamagnetic; Fe (Iron), and Gd (Gadolinium) are ferromagnetic; Cu^{2+} (Cupric ion) and Cr^{3+} (Chromic ion) are paramagnetic; while MnO (Manganese oxide) is antiferromagnetic.
Hence, option (b) is the correct answer.

8. Which one of the following pairs of metals constitutes the lightest metal and the heaviest metal, respectively?

(a) Lithium and Mercury

(b) Lithium and Osmium

(c) Aluminium and Osmium

(d) Aluminium and Mercury

I.A.S. (Pre) 2008

Ans. (b)

The atomic weight and density of given metals are as follows-			
Metals	**Atomic No.**	**Atomic Weight (amu)**	**Density (g/cc)**
1. Lithium	3	6.941	0.534
2. Mercury	80	200.59	13.534
3. Osmium	76	190.23	22.61
4. Aluminium	13	26.982	2.70

From the above mentioned data, it is clear that Lithium is the lightest and Osmium is the heaviest metal. Osmium is the densest naturally occurring metal. Therefore, it is the heaviest metal.

9. Which one of the following is the hardest metal?

(a) Gold (b) Iron

(c) Platinum (d) Tungsten

U.P.P.C.S. (Pre) 1996

Ans. (d)

Among all natural materials, diamond is the hardest (which is non-metal) whereas, among the metals, Tungsten is the hardest.

10. Which is the hardest in the following?

(a) Diamond (b) Glass

(c) Quartz (d) Platinum

44th B.P.S.C. (Pre) 2000

M.P.P.C.S. (Pre) 1992

Ans. (a)

Diamond is the hardest, least compressible and best thermal conductor among all natural materials.

11. Which of the following is a naturally occurring hardest substance on the Earth?

(a) Graphite (b) Wurtzite boron nitride

(c) Iron (d) Diamond

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (d)

Diamond is the hardest naturally occurring substance present on the Earth. However, wurtzite boron nitride is believed to be harder than diamond. It has a similar structure to diamond but is made up of different atoms. It is formed during volcanic eruptions with high temperature and pressure. The simulations showed that wurtzite boron nitride could handle 18% more stress than diamond, which is due to the re-orientation of the flexible bonds and the more complex structure than diamonds. But because only minute amounts of this mineral have been discovered, its hardness properties are yet to be experimentally tested. That's why UPPSC experts had answered option (d) for this question in their official answer key.

12. Which one of the following materials is very hard and very ductile ?

(a) Carborundum (b) Tungsten

(c) Cast iron (d) Nichrome

I.A.S. (Pre) 2000

Ans. (d)

Nichrome (NiCr, nickel-chrome, chrome-nickel, etc.) generally refers to any alloy of Nickel, Chromium and often Iron and/or other elements or substances. Nichrome is very hard and very ductile material. It has high specific resistivity, high melting point and minimum temperature coefficient. It also has the ability to operate at high temperature. Nichrome alloys are typically used in resistance wire. They are also used in some dental restorations (fillings) and in other applications.

13. The heaviest natural element is ?

(a) Uranium (b) Mercury

(c) Gold (d) Calcium

U.P. Lower Sub. (Pre) 2004

Ans. (a)

Uranium is the heaviest naturally occurring element with an atomic no. of 92. It is a pure form of silver-coloured heavy metal. Its most common isotope Uranium-238 has a nucleus containing 92 protons and 146 neutrons. It has a density of 19.05 g/cm^3.

14. Heaviest metal of the following is made of –

(a) Copper (b) Uranium

(c) Aluminium (d) Silver

40th B.P.S.C. (Pre) 1995

Ans. (b)

See the explanation of above question.

15. The heaviest metal among the following is :

(a) Gold (b) Silver

(c) Mercury (d) Platinum

(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

Among the given options, platinum is the heaviest metal with a density of 21.09 g/cm^3.

16. The costliest metal of the world discovered recently is :

(a) Endohedral Fullerene (b) Californium 252

(c) Tritium (d) Rhodium

U.P. P.C.S. (Pre) 2016

Ans. (*)

The material, there discovery and approx. cost per gram (as per the question period) in US$ are given below–		
Material	**Discovery (in year)**	**Cost per gram (in US $)**
Tritium	1934	30,000
Rhodium	1803	58
Californium-252	1950	27 million
Endohedral Fullerene	1985	167 million

From the above list, it is clear that Endohedral Fullerene is the costliest material of the world discovered recently. But as the question asked about the costliest metal, it is not a correct answer. However, there are two types of Endohedral Fullerense namely – Endohedral metallofullerenes and non-metal doped fullerene. Californium-252 is the costliest metal but it is not discovered recently. So the question seems to be incorrect.

Note : As of 1 April, 2025, Rhodium is priced at approx. $ 206 per gram.

17. The chemical structure of the pearl is –

(a) Calcium Carbonate
(b) Calcium Carbonate & Magnesium Carbonate
(c) Calcium Chloride
(d) Calcium Sulphate

Uttarakhand P.C.S. (Pre) 2007

Ans. (a)

The chemical composition of pearl is about 82 – 86% Calcium Carbonate (Aragonite), 10–14% Conchiolin and 2–4% of water ($CaCO_3$ and H_2O). Conchiolin is a protein and a binding agent.

18. The main constituents of pearl are –

(a) Calcium Carbonate and Magnesium Carbonate
(b) Aragonite and Conchiolin
(c) Ammonium sulphate and Sodium Carbonate
(d) Calcium oxide and Ammonium Chloride

I.A.S. (Pre) 1994

Ans. (b)

See the explanation of above question.

19. Pearl is mainly constituted of –

(a) Calcium Oxalate (b) Calcium Sulphate
(c) Calcium Carbonate (d) Calcium Oxide

R.A.S./R.T.S.(Pre) 2008

Ans. (c)

See the explanation of above question.

20. What are Rubies and Sapphires chemically known as?

(a) Silicon Dioxide (b) Aluminium Oxide
(c) Lead Tetroxide (d) Boron Nitride

I.A.S. (Pre) 2008

Ans. (b)

Aluminium Oxide is a chemical compound of Aluminium and Oxygen with the chemical formula Al_2O_3. Corundum is the most common naturally occurring crystalline form of Aluminium Oxide. Rubies and Sapphires are gem-quality in forms of Corundum which owe their characteristic colours to trace impurities.

21. The chemical formula of sapphire (Ruby) is :

(a) Al_2O_3 (b) Al_3O_2
(c) N_2O (d) NO_2
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

The chemical formula of sapphire (Ruby) is Al_2O_3 (Aluminium Oxide). It is a precious gemstone, a variety of mineral corundum.

22. Assertion (A) : Sodium metal is stored under kerosene.
Reason (R) : Metallic sodium melts when exposed to air.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not a correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

I.A.S. (Pre) 1998

Ans. (c)

Sodium is kept in kerosene to prevent it from coming in contact with oxygen present in the air. If this happens, it will react with the oxygen and form sodium oxide. This is a strongly exothermic reaction and a lot of heat is generated. Thus, Sodium is kept under kerosene.

23. Which one of the following elements is kept safely in Kerosene oil?

(a) Sodium (b) Copper
(c) Mercury (d) Silver

U.P.P.C.S. (Mains) 2017

Ans. (a)

See the explanation of above question.

24. These days yellow lamps are frequently used as street light. Which one of the following is used in these lamps :

(a) Sodium (b) Neon
(c) Hydrogen (d) Nitrogen

U.P.P.C.S. (Pre) 2000

Ans. (a)

There are 2 types of street light lamps which are used by municipalities. They are sodium vapor and mercury vapor lamps. The mercury vapor lamps have usually a white ambient light and sodium vapor lamps have orange/yellow light. Compared to LPS (Low-pressure sodium) lamps, high-pressure sodium lamps tend to have a longer life, less lumen per watt efficiency and most importantly a higher colour rendering index.

25. Sodium Vapor Lamp is usually used as street light, because –

(a) These are cheap.
(b) Light from this is monochromatic and will not split through water droplets.
(c) It is pleasing to the eyes.
(d) It is brightly illuminating.

U.P.P.C.S. (Pre) 2007

Ans. (b)

A sodium-vapor lamp is a gas-discharge lamp that uses sodium in an excited state to produce light. Low-pressure sodium lamps only give monochromatic yellow light and so inhibit colour vision at night and will not split through water droplets. Sodium-vapor lamps cause less light pollution than mercury-vapor lamps.

26. Which one of the following metals is accessed in the native state ?

(a) Aluminium (b) Gold
(c) Chromium (d) Zinc

U.P.P.C.S. (Mains) 2016

Ans. (b)

Among the given metals, gold is the least reactive, so this was accessed in the native (free) state. Only metals like gold, silver, and platinum etc. occur in native state in nature in large amounts.

27. Consider the following statements with respect to noble metals :

1. Noble metals are found in pure form in nature.
2. Uranium and lead are examples of noble metal.
Which of the above statements is/are correct?

(a) 2 only
(b) Both 1 and 2
(c) Brass is also noble metal
(d) 1 only
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

A noble metal is ordinarily regarded as a metallic chemical element that is generally resistant to corrosion and is usually found in nature in its raw form. Gold, platinum, and the other platinum group metals (ruthenium, rhodium, palladium, osmium, iridium) are most often so classified. Silver, copper and mercury are sometimes included as noble metals, however less often as each of these usually occurs in nature combined with sulfur. Uranium and lead are not examples of noble metals.

28. Gold is dissolved in –

(a) Sulphuric acid
(b) Nitric acid
(c) Mixture of Sulphuric and Nitric Acid
(d) Hydrochloric acid

47th B.P.S.C. (Pre) 2005

Ans. (*)

Gold is unaffected in air, water, alkali halogen and all acids except Aqua regia (a mixture of hydrochloric acid and nitric Acid in a 3:1 ratio). The name Aqua-regia was coined by chemists because of its ability to dissolve gold "the king of metals". It is a mixture of acids, a fuming yellow or red solution.

29. Of how many carats is the pure gold?

(a) 22 (b) 24
(c) 28 (d) 20

M.P.P.C.S. (Pre) 1995

Ans. (b)

It is an extension of the older carat (Karat in American spelling) system of denoting the purity of gold by fractions of 24, such as '18 carat' for an alloy with 75% (18 parts per 24) pure gold by mass. Because of the softness of pure (24 carat) gold, it is usually alloyed with base metals for use in jewellery, altering its hardness and ductility, melting point, colour and other properties. Alloys with lower carat rating typically 22k, 18k, 14k or 10k contain higher percentages of copper or other base metals or silver or palladium in the alloy.

30. Which among the following is also known as white metal?

(a) Nickel
(b) Rhodium
(c) Platinum
(d) Palladium
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (e)

In its most basic form, white metal is an alloy of lead and lithium, or other metals like cadmium, bismuth, and zinc. The term 'white metals' is generally used to describe a series of metal alloys with decorative bright and are mostly used as a base for plated silverware, ornaments, or novelties. The term is also used in the antiques trade for an item suspected of being silver, but not hallmarked. It is to be noted that platinum is a naturally occuring white metal.

31. Minerals are :

(a) Liquids (b) Inorganic solids
(c) Gases (d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

According to Geologists, minerals are naturally occurring inorganic substances with a definite and predictable chemical composition and physical properties.

32. Ilmenite, which is widely distributed along the Indian coastline, is a mineral of :

(a) tungsten (b) titanium
(c) gallium (d) tin
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

Ilmenite is a titanium-iron oxide mineral with the idealized formula $FeTiO_3$. It is a weakly magnetic black or steel-gray solid. Ilmenite is the most important ore of titanium and the main source of titanium dioxide, which is used in paints, printing inks, fabrics, plastics, paper, sunscreen, food and cosmetics. Ilmenite and rutile (TiO_2-also a mineral of ilmenite) along with other heavy minerals are important constituents of beach sand deposits found right from Moti Daman-Umbrat coast (Gujarat) in the west to Odisha coast in the east.

33. The softest mineral, Talc (Soapstone) is mainly :

(a) Manganese Silicate (b) Sodium Silicate
(c) Sodium Phosphate (d) Magnesium Silicate

R.A.S./R.T.S.(Pre) 1999

Ans. (d)

Talc is a mineral which is composed of hydrated Magnesium silicate with formula $Mg_3Si_4O_{10}(OH)_2$. On the Mohs hardness scale, the softest mineral talc is rated 1 and the hardest mineral, the diamond is rated 10. In loose form, talc is the widely used substance known as a baby powder (aka talcum). It occurs as foliated to fibrous masses and in an exceptionally rare crystal form.

34. Which among the following is known as quicklime?

(a) CaO (b) $CaCO_2$
(c) $Ca(OH)_2$ (d) $CaCl_2$
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Calcium oxide (CaO), commonly known as quicklime or burnt lime, is a widely used chemical compound. It is a white, caustic, alkaline, crystalline solid at room temperature.

35. The chemical name of limestone is?

(a) Calcium Carbonate (b) Magnesium Chloride
(c) Sodium Chloride (d) Sodium Sulphide

U.P.P.C.S. (Pre) 1993

Ans. (a)

Limestone is a sedimentary rock composed largely of the minerals Calcite and Aragonite which are different crystal forms of Calcium Carbonate ($CaCO_3$). Limestone binds with silica and other impurities to remove them from the iron.

36. Which of the following is the main ingredient of cement?

(a) Limestone (b) Silica clay
(c) Gypsum (d) Ash
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (e)

Cement is made with limestone, silica clay, and marl as the main raw materials. Lime is the main ingredient of cement, accounting for about 60 – 65% of total cement weight. Silica is the second major ingredient of cement and about 17 to 25% of cement ingredients is silica. Cement (OPC) is made by heating limestone (calcium carbonate) with other materials (such as clay) to 1,450 °C (2,640 °F) in a kiln, in a process known as calcination that liberates a molecule of carbon dioxide from the calcium carbonate to form calcium oxide (CaO), or quicklime, which then chemically combines with the other materials in the mix to form calcium silicates and other cementitious compounds. The resulting hard substance, called 'clinker', is then ground with a small amount of gypsum ($CaSO_4.2H_2O$) into a powder to make ordinary Portland cement, the most commonly used type of cement (often referred to as OPC).

37. 'Plaster of Paris' is made up of :

(a) Marble (b) Cement
(c) Gypsum (d) Limestone

Chhattisgarh P.C.S. (Pre) 2019

Ans. (c)

Plaster of Paris is a white powdery slightly hydrated calcium sulfate ($CaSO_4.1/2\ H_2O$) made by calcining gypsum and used mainly for casts and molds in the form of a quick-setting paste with water.

38. To protect broken bones, Plaster of Paris is used. It is :

(a) Slaked lime (b) Calcium carbonate
(c) Calcium oxide (d) Gypsum
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

See the explanation of above question.

39. Chemically 'Plaster of Paris' is :

(a) Calcium Sulphate (b) Calcium Carbonate
(c) Calcium Oxide (d) Calcium Oxalate

R.A.S./R.T.S.(Pre) 2007

Ans. (a)

See the explanation of above question.

40. The chemical formula of the Plaster of Paris is –

(a) $CaSO_4$ (b) $CaSO_4\ ½\ H_2O$
(c) $CaSO_4.H_2O$ (d) $CaSO_4.2H_2O$

42nd B.P.S.C. (Pre) 1997
39th B.P.S.C. (Pre) 1994

Ans. (b)

See the explanation of above question.

41. Doctors, Artists and Sculptors use Calcium Sulphate which is popularly known as –

(a) Quick lime (b) Limestone
(c) Bleaching powder (d) Plaster of Paris

U.P. Lower Sub. (Spl.) (Pre) 2004

Ans. (d)

See the explanation of above question.

42. Which one of the following materials contains calcium?

(a) China clay (b) Corundum
(c) Gypsum (d) Talc

U.P.P.C.S. (Pre) 2019

Ans. (c)

Gypsum is a soft sulfate mineral which contains calcium. It is composed of calcium sulphate dihydrate, with the chemical formula $CaSO_4.2H_2O$. It is widely mined and is used as a fertilizer and as the main constituent in many forms of plaster, blackboard/sidewalk chalk, and drywall.

43. Monazite is an ore of –

(a) Zirconium (b) Thorium
(c) Titanium (d) Iron

I.A.S. (Pre) 1994

Ans. (b)

Monazite is an important ore for Thorium, Lanthanum and Cerium. India, Madagascar and South Africa have large deposits of monazite sands. The deposits in India are particularly rich in Monazite. Its deposits are found in coastal beach placer sands in parts of Kerala, Tamil Nadu, Odisha, Andhra Pradesh, Maharashtra and Gujarat and in the inland placers in parts of Jharkhand, West Bengal and Tamil Nadu.

44. Mica is a :

(a) Good conductor of heat and bad conductor of electricity
(b) Bad conductor of both heat and electricity
(c) Good conductor of heat and electricity both
(d) Bad conductor of heat and good conductor of electricity

Chhattisgarh P.C.S. (Pre) 2003

Ans. (a)

(1) Good conductors of electricity are : Aluminum, Brass, Copper, Iron, Steel (2) Bad conductors of electricity are : Acrylic, China clay, Glass, Mica, Paper, Plastic & Wood. Mica is a bad conductor of electricity but a good conductor of heat. It is used as a raw material in electrical industry.

45. In which of the following industries is mica used as a raw material –

(a) Iron and steel (b) Toys
(c) Glass and pottery (d) Electrical

U.P.P.C.S (Pre) 2010

Ans. (d)

See the explanation of above question.

46. Which among the following liquids is the best conductor of heat?

(a) Mercury (b) Water
(c) Ether (d) Benzene

U.P.P.C.S. (Pre) 2005
U.P.P.C.S. (Mains) 2014

Ans. (a)

Mercury is a chemical element with the symbol (Hg). Mercury is used in thermometers due to its special properties. It can measure a wide range of temperatures from -40 to 356°C and up to 570°C under pressure in a liquid state. It expands regularly in proportional to the absolute temperature changes. A heavy, silvery d-block element, mercury is the only common metal element which is found in liquid state at standard conditions for temperature and pressure. It is a metal, so it is a conductor of heat. Mercury is the poorest conductor of heat among all metals, but among the given options it is the best conductor of heat.

47. The poorest conductor of heat among the following is :

(a) copper (b) lead
(c) mercury (d) zinc
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

All metals are conductors of heat but lead and mercury are poor conductors of heat compared to other metals. Thermal conductivity of lead is about 34.7 w/mK while thermal conductivity of mercury is only about 8.31 w/mK. Hence, mercury is the poorest conductor of heat among all metals.

48. Which one of the following metals is liquid at ordinary temperature?

(a) Lead (b) Nickel
(c) Mercury (d) Tin

U.P.P.C.S. (Mains) 2014
44th B.P.S.C. (Pre) 2000

Ans. (c)

Mercury is the only common metal which is liquid at ordinary (room) temperature. Mercury is sometimes called quicksilver.

49. Which of the following is in liquid form at room temperature?

(a) Lithium (b) Sodium
(c) Francium (d) Cerium

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

There are two elements Bromine and Mercury that are liquid at the room temperature (298 K or 25°C). There are four more elements which melt just a few degree above room temperature. These are Francium, Caesium, Gallium and Rubidium.

50. Mercury is basically used in thermometer devices because its especiality is –

(a) High density
(b) High liquidity
(c) High circulation power
(d) High specific heat

U.P. Lower Sub. (Pre) 2003

Ans. (c)

Mercury is easily the best liquid to use in thermometers, five important reasons are –

(1) It is very reflective, so it's easy to see and to read accurately.
(2) It doesn't wet the glass, so you don't get an inaccurate reading if the temperature is falling.
(3) It is a metal, so it is a conductor of heat.
(4) It expands evenly with the temperature with high circulation power, so a linear scale can be used with a high degree of accuracy.
(5) There is a large range of temperature for which it is a liquid.

51. Which of the following do not react with water at all?

(a) Iron (b) Lead
(c) Magnesium (d) Aluminium
(e) None of the above

Chhattisgarh P.C.S. (Pre) 2015

Ans. (b)

Metals like potassium and sodium react violently with cold water. The reaction of calcium with water is less violent. Magnesium does not react with cold water but it reacts with hot water. Metals like aluminium, iron and zinc do not react either with cold or hot water but they react with steam. Metals such as lead, copper, silver, gold and platinum do not react with water at all (neither with cold or hot water nor with steam).

52. There is no reaction when steam passes over :

(a) Aluminium (b) Copper
(c) Carbon (d) Iron

38th B.P.S.C. (Pre) 1992

Ans. (b)

See the explanation of above question.

53. Iron is obtained from :

(a) Limestone (b) Pitch-blende
(c) Monazite Sand (d) Haematite

R.A.S./R.T.S.(Pre) 1999

Ans. (d)

Iron ores are rocks and minerals from which metallic iron can be economically extracted. The ores are usually rich in iron oxides and the iron is usually found in the form of magnetite (Fe_3O_4 – 72.4 % Fe) and haematite (Fe_2O_3 – 69.9 % Fe).

54. Which of the following iron ores is mined at Bailadila?

(a) Haematite (b) Siderite

(c) Limonite (d) Magnetite
(e) None of the above / More than one of the above

60th- 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Bailadila range of mines is perched on the southern tip of Chhattisgarh in Dantewada district. Very high grade iron ore haematite is mined at Bailadila. The association of very rich and extensive iron ores with haematite in the Bailadila range has first been made known to the world between 1898-1900 by P.N. Bose, who was the first to do geological mapping of this region.

55. In the case of rusting, the weight of iron –

(a) Increases (b) Decreases
(c) Remains the same (d) Uncertain

U.P.P.C.S. (Mains) 2008
M.P.P.C.S. (Pre) 1991

Ans. (a)

Due to rust, the weight of iron increases as iron is converted into iron oxide after chemical reaction with oxygen, in presence of humidity.

56. What happens to the weight of iron, when it rusts?

(a) Increases for long time
(b) Decreases then increases
(c) Increases then decreases
(d) Remains same
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

See the explanation of above question.

57. Consider these statements and choose the right answer from the given code :

Statement (A) : Normally it has been seen that iron goods covered with a brown powder called rust when they are unsheltered in the atmosphere.

Statement (R) : Rust or the brown powder is the result of deposition of tannin.

Code :

(a) Both (A) and (R) is correct, and (R) is right explanation of (A).
(b) Both (A) and (R) is correct, but (R) is not the right explanation of (A).
(c) (A) is correct, but (R) is wrong.
(d) (A) is wrong, but (R) is correct.

U.P. Lower Sub. (Pre) 1998

Ans. (c)

Statement (A) is correct but statement (R) is wrong because rust is iron oxide (not tannin), a usually red oxide formed by the redox reaction of iron and oxygen in the presence of water or air moisture. Tannin is a pale-yellow to light-brown substance secreted from plants and used chiefly in tanning leather, dyeing fabric, making ink, and in various medical applications.

58. Statement (A) : Galvanized iron does not rust.
Statement (R) : Zinc has the efficiency of oxidation.
Code :

(a) Both (A) and (R) is correct, and (R) is right clarification of (A).
(b) Both (A) and (R) is correct, but (R) is not the right classification of (A).
(c) (A) is correct, but (R) is wrong.
(d) (A) is wrong, but (R) is correct.

U.P. Lower Sub. (Pre) 2002

Ans. (a)

Galvanization is the process of applying a protective zinc coating on steel or iron to prevent them from rusting and oxidation. The zinc forms a barrier between atmospheric oxygen and the underlying iron or steel. It does this by transferring electrons and oxidizing more quickly than iron. This rapid zinc oxidation prevents ferrous metals from rusting until the zinc has exhausted its free-electron capacity or the protective coating has worn away.

59. Galvanized iron pipes have a coating of :

(a) Zinc (b) Mercury
(c) Lead (d) Chromium
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

See the explanation of above question.

60. The plates of galvanized iron remains protected from rust because the existence of –

(a) Lead (b) Chromium
(c) Zinc (d) Bung

I.A.S. (Pre) 1994

Ans. (c)

See the explanation of above question.

61. Galvanized iron is coated with –

(a) Aluminium (b) Galena
(c) Silver (d) Zinc

Chhattisgarh P.C.S. (Pre) 2011

Ans. (d)

See the explanation of above question.

62. **Which one of the following is essential in corrosion of iron metal?**
(a) Oxygen only (b) Oxygen and moisture
(c) Hydrogen only (d) Hydrogen and moisture
U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (b)

Oxygen and moisture are essential in corrosion of iron metal. Iron corrosion is generally characterized by the formation of rust due to an electrochemical process in the presence of oxygen and moisture (water) in the surrounding environment. When iron reacts with water and oxygen, iron (II) hydroxide is formed. The latter further reacts with oxygen and water to form hydrated iron (III) oxide-widely known as rust.

63. **Which metal is generally used for coating of brass utensils to prevent copper contamination?**
(a) Tin (b) Zinc
(c) Aluminium (d) Lead
R.A.S/R.T.S (Pre) 2018

Ans. (a)

The coating of tin on brass utensils prevents copper contamination. Tin is a soft and white metal like silver. Its symbol is Sn with atomic no. 50.

64. **Aluminium surface is often 'Anodized'. This means the deposition of a layer of –**
(a) Chromium Oxide (b) Aluminium Oxide
(c) Nickel Oxide (d) Zinc Oxide
I.A.S. (Pre) 2000

Ans. (b)

Anodizing is an electrochemical process by which the surface of a metal is made durable and rust resistant. In this process, a layer of aluminium oxide is deposited on aluminium.

65. **Standard electrode potential of three metals A, B and C are respectively 0.5 V, – 3 V and – 1.2 V. Reducing power of these metals would be :**
(a) A > C > B (b) B > C > A
(c) C > B > A (d) None of the above
70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (b)

The more negative standard electrode potential of a metal, the greater its tendency to be oxidised (lose electrons) and act as a stronger reducing agent.

Metal	**Standard Electrode Potential**
A	0.5 V
B	– 3 V
C	– 1.2 V

Hence, Reducing Power (Oxidising Ability) : B > C > A

66. **Match List-I with List-II and select the correct answer from the code given below the lists:**

List-I (Metal)	**List-II (Property)**
A. Sodium	**1. Good conductor of electricity**
B. Mercury	**2. Liquid at room temperature**
C. Silver	**3. Poor conductor of heat**
D. Lead	**4. Can be easily cut with knife**

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	1	4	3	2
(c)	4	2	1	3
(d)	4	1	2	3

U.P. P.C.S. (Pre) 2020

Ans. (c)

The correctly matched lists are as follows :

List-I (Metal)		**List-II (Property)**
Sodium	–	Can be easily cut with knife
Mercury	–	Liquid at room temperature
Silver	–	Good conductor of electricity
Lead	–	Poor conductor of heat

67. **Match List-I with List-II and select the correct answer using the codes given below the lists :**

List-I	**List-II**
A. Best conductor of heat and electricity	**1. Gold**
B. Metal found in highest amount	**2. Lead**
C. Most flexible metal and able to increase by bang	**3. Aluminium**
D. Minimum heat conducting	**4. Silver**

Codes :

	A	B	C	D
(a)	1	3	2	4
(b)	2	3	4	1
(c)	3	2	4	1
(d)	4	3	1	2

U.P. Lower Sub. (Pre) 2002

Ans. (d)

The correctly matched lists are as follows :
Best conductor of heat and electricity – Silver
Metal found in highest amount – Aluminium
Most flexible metal and able to increase by bang – Gold
Minimum heat conducting metal – Lead

68. **Which of the following materials has the highest electrical conductivity ?**
(a) Diamond (b) Silver
(c) Graphite (d) Wood
Uttrakhand U.D.A./L.D.A. (Mains) 2006
Ans. (b)

See the explanation of above question.

69. **Which of the following is mainly used for the production of Aluminium?**
(a) Haematite (b) Lignite
(c) Bauxite (d) Magnetite
M.P.P.C.S. (Pre) 2005
Ans. (c)

Bauxite is an ore of aluminium, which is found in the form of hydrated aluminium oxides. It is used as a principal raw material in aluminium industry. It consists mostly of the minerals gibbsite $Al(OH)_3$, boehmite γ-AlO(OH) and diaspore α-AlO(OH) mixed with the two iron oxides goethite and haematite. The French geologist first discovered bauxite near the village of Les Baux, Southern France.

70. **The most important ore of aluminium :**
(a) bauxite (b) calamine
(c) calcite (d) galena
(e) None of the above/More than one of the above
67th B.P.S.C. (Pre) 2022
Ans. (a)

See the explanation of above question.

71. **Bauxite, is an ore of which of the following metals?**
(a) Iron (b) Copper
(c) Aluminium (d) Silver
Uttarakhand P.C.S. (Pre) 2007
U.P. U.D.A./L.D.A. (Pre) 2006
Ans. (c)

See the explanation of above question.

72. **Bauxite is the ore of :**
(a) Iron (b) Aluminium
(c) Copper (d) Gold
U.P.P.C.S. (Pre) (Re. Exam) 2015
Ans. (b)

See the explanation of above question.

73. **Aluminium metal is obtained from :**
(a) Pitch blende (b) Graphite
(c) Bauxite (d) Argentite
Uttrakhand U.D.A./L.D.A. (Mains) 2007
Ans. (c)

See the explanation of above question.

74. **Which of the following industries uses bauxite as a principal raw material?**
(a) Aluminium (b) Cement
(c) Fertilizer (d) Ferro-manganese
Chhattisgarh P.C.S. (Pre) 2018
Ans. (a)

See the explanation of above question.

75. **Which of the following corundum and cryolite are important ores ?**
(a) Aluminium (b) Silver
(c) Tin (d) Iron
70th B.P.S.C. (Pre) 2024
Ans. (a)

Bauxite ($Al_2O_3.2H_2O$), Cryolite (Na_3AlF_6), Corundum (Al_2O_3) and Diaspore [AlO(OH)] are major ores of aluminium. Aluminium is found on Earth primarily in rocks in the crust, where it is the third-most abundant element, after oxygen and silicon.

76. **Which one of the following statements is correct ?**
(a) Liquid sodium is employed as a coolant in nuclear reactors
(b) Calcium carbonate is an ingredient of toothpaste
(c) Bordeaux mixture consists of sodium sulphate and lime
(d) Zinc amalgams are used as a dental filling
I.A.S. (Pre) 2003
Ans. (b)

The precipitated calcium carbonates (PCCs) and ground calcium carbonates (GCCs) are used for general purpose toothpaste specially dentifrices and other oral care products. Calcium carbonate is used as an abrasive in toothpaste. Calcium carbonate is insoluble in water so it can only be used in opaque products, not in clear gels.
Generally normal water or heavy water is used as coolant under high pressure in nuclear reactors. Molten sodium is used as coolant in Fast Breeder Test Reactor (FBTR). Bordeaux mixture consists of copper sulphate and lime.
Zinc amalgams are not used as a dental filing. Dental amalgam is produced by mixing liquid mercury with an alloy made of silver, tin, and copper solid particles. Small quantity of zinc, mercury and other metals may be present in some alloys.

77. **Match List-I (Industrial process) with List-II (Industry with which associated) and select the correct answer using the codes given below the lists.**

List-I	List-II
A. Cracking	**1. Rubber**
B. Smelting	**2. Petroleum**
C. Hydrogenation	**3. Copper**
D. Vulcanization	**4. Edible Fats**

Code :

	A	B	C	D
(a)	3	2	1	4
(b)	2	3	4	1
(c)	2	3	1	4
(d)	3	2	4	1

I.A.S. (Pre) 2000

Ans. (b)

Cracking	–	Petroleum
Smelting	–	Copper
Hydrogenation	–	Edible Fats
Vulcanization	–	Rubber

78. Match List-I with List-II and select the correct answer using the codes given below the lists :

List - I (Naturally occurring substance)	List-II (Elements present)
A. Diamond	**1. Calcium**
B. Marble	**2. Silicon**
C. Sand	**3. Aluminium**
D. Ruby	**4. Carbon**

Code :

	A	B	C	D
(a)	3	1	2	4
(b)	4	2	1	3
(c)	2	1	3	4
(d)	4	1	2	3

I.A.S. (Pre) 1999
U.P.P.C.S. (Pre) 2010
U.P.U.D.A./L.D.A. (Pre) 2010

Ans. (d)

Diamond is a metastable allotrope of carbon where the carbon atoms are arranged in a variation of the face-centered cubic crystal structure called a diamond lattice. Marble is a metamorphic rock composed mainly of crystalline calcium carbonate or calcium magnesium carbonate. The most common component of sand is silicon dioxide in the form of quartz. Ruby is considered as one of the four precious stones together with sapphire, emerald and diamond. The main components of ruby are aluminium, oxygen and chromium.

79. Which one of the following pairs is not correctly matched?

(a) Aluminium – Bauxite (b) Copper – Cinnabar
(c) Zinc – Calamine (d) Iron – Haematite

U.P. R.O./A.R.O. (Pre) 2017

Ans. (b)

The correctly matched lists are as follows :

Aluminium	–	Bauxite
Copper	–	Cuprite
Mercury	–	Cinnabar
Zinc	–	Calamine (Smithsonite & Hemimorphite)
Iron	–	Haematite

80. Which of the following is lighter than water?

(a) Aluminium (b) Sodium
(c) Magnesium (d) Manganese

Uttarakhand U.D.A. /L.D.A. (Pre) 2003

Ans. (b)

Water has a defined density (1 gram per cubic centimetre) while the lightest metals are lighter than water. They are Lithium 0.53 gm/cm^3, Potassium 0.862 g/cm^3 and Sodium 0.971 g/cm^3. These are malleable and highly reactive so they are impractical to use as the basis of an alloy with any structural utility.

81. Which one of the following is the correct sequence of the given substances in the decreasing order of their densities?

(a) Steel > Mercury > Gold
(b) Gold > Mercury > Steel
(c) Steel > Gold > Mercury
(d) Gold > Steel > Mercury

I.A.S. (Pre) 2005

Ans. (b)

Substance	**Density (gram/cm^3)**
Gold	19.3
Mercury	13.6
Steel	7.8

Thus, the correct sequence of the given substances in the decreasing order of their densities is as follows :
Gold > Mercury > Steel.

82. Which of the following has highest melting point?

(a) Boron (b) Iron
(c) Silicon (d) Aluminium